MAKERERE UNIVERSITY | COLLEGE OF COMPUTING & INFORMATION SCIENCES

FULL DISTRIBUTED SYSTEM ARCHITECTURE SPECIFICATION

Multi-Tier Hardware, FreeRTOS Concurrency & ESP-NOW Protocols

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FULL DISTRIBUTED SYSTEM ARCHITECTURE

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Role: Group Leader, System Architect, and Integrator

Group: SE-24 | Makerere University (CoCIS) - BSE III & CS II

Course: Embedded Systems Coursework (Academic Year 2026/2027)

Project: Smart Control and Monitoring System for CoCIS

Communication Bus: Wireless ESP-NOW (2.4 GHz Peer-to-Peer Wi-Fi)

Node Designations: ENTERANCE_MASTER, ROOM_NODE_1, ROOM_NODE_2

Associated Diagram: SYSTEM_ARCHITECTURE_DIAGRAM.svg

1. Architectural Philosophy & Overview

The CoCIS Smart Control and Monitoring System is engineered as a distributed, multi-tier embedded IoT architecture. Rather than running single-point monolithic wiring across campus facilities, the design partitions responsibilities across three autonomous ESP32 microcontrollers communicating wirelessly via 2.4 GHz ESP-NOW:

- 1. Central Gateway & Access Terminal (ENTERANCE_MASTER - Master ESP32):
- Controls physical building ingress and egress with dual RFID readers and servo barriers.
- Enforces building security policies: configurable schedule curfew, anti-passback, and intrusion lockout.
- Serves as the single authority for non-volatile SD storage (FATFS), preventing file access conflicts.
- Hosts the administrator Serial CLI management console over USB-UART0.
- Operates a FreeRTOS real-time synchronization engine with priority queues and mutexes.
- 2. Autonomous Lecture Room Nodes (ROOM_NODE_1 & ROOM_NODE_2 - Edge ESP32s):
- Continuously evaluate local occupancy using PIR motion sensors and ambient light sensors (LDR).
- Automatically control lecture room lighting based on motion and daylight availability.
- Stream telemetry (occupancy, light relay state, lux level, energy metrics) wirelessly to ENTERANCE_MASTER every 2 seconds.
- Respond immediately to Master broadcast commands (such as campus-wide emergency evacuations).

2. Full System Architecture Diagram (Mermaid)

```
flowchart TD
    subgraph EntranceMaster["ENTERANCE_MASTER: Entrance Master & Gateway (ESP32 #1)"]
        direction TB
        subgraph PhysicalEntry["Physical Entry Section"]
            RFID_IN["RFID Entry Reader\n(RC522 - VSPI)"]
            SERVO_IN["Entry Door Servo\n(GPIO 13)"]
            LED_GREEN["Access Granted LED\n(GPIO 2)"]
```

```
        end

        subgraph PhysicalExit["Physical Exit Section"]
            RFID_OUT["RFID Exit Reader\n(RC522 - Shared VSPI)"]
            SERVO_OUT["Exit Door Servo\n(GPIO 15)"]
            LED_RED["Access Denied LED\n(GPIO 4)"]
            BUZZER["Curfew/Alarm Buzzer\n(GPIO 12)"]
        end

        subgraph FreeRTOSCore["ESP32 Dual-Core RTOS & Concurrency Engine"]
            direction TB
            subgraph Tasks["FreeRTOS Tasks"]
                T_ACCESS["AccessControlTask (Core 1)"]
                T_LOG["SDLoggerTask (Core 0)"]
                T_WIFI["WirelessRxTask (Core 1)"]
                T_CLI["SerialCLITask (Core 0)"]
                T_HB["HeartbeatMonitor (Core 1)"]
            end
            subgraph Queues["IPC Queues"]
                Q_ACCESS["xAccessEventQueue"]
                Q_LOG["xSDLogQueue"]
                Q_WIFI["xWirelessPacketQueue"]
            end
            subgraph Mutexes["Thread-Safe Mutexes"]
                M_OCC["xOccupancyMutex"]
                M_SCHED["xScheduleMutex"]
                M_SD["xSDCardMutex"]
                M_ROOM["xRoomStateMutex"]
            end
        end

        subgraph StorageSubsystem["Storage Subsystem (MicroSD SPI)"]
            F_USERS["users.csv\n(Tags & Roles)"]
            F_LOGS["access_log.csv\n(Audit Trail)"]
            F_OCC["occupancy.csv\n(Crash Recovery)"]
            F_CONF["config.txt\n(Schedule & Lockout)"]
        end

        subgraph ManagementConsole["Serial CLI Interface"]
            CLI["USB-UART0 Console\n(115200 Baud)"]
            AUTH["SHA-256 Login &\nSession Manager"]
        end
    end

    subgraph WirelessBus["2.4 GHz ESP-NOW Wireless Protocol Bus (Connectionless Peer-to-Peer)"]
        direction TB
        WIFI_TRANS["RF Transceiver (WIFI_MODE_STA)\nSub-5ms Latency | Hardware MAC ACKs | No Ro
        PKT_TELEMETRY["Telemetry Packets: occupied, lights_on, lux, energy_sec"]
        PKT_EMERGENCY["Broadcast Alert: Instant Evacuation (0xFF:FF:FF:FF:FF:FF)"]
    end

    subgraph RoomNode1["ROOM_NODE_1: Lecture Room 1 (ESP32 #2)"]
        PIR_1["PIR Motion Sensor\n(GPIO 14 - Interrupt)"]
        LDR_1["Ambient Light LDR\n(GPIO 34 - ADC1)"]
        RELAY_1["Lighting Relay / LED\n(GPIO 12)"]
        HB_LED_1["Heartbeat LED\n(GPIO 2)"]
        CTRL_1["Energy Logic: 30s Countdown\n+ Daylight Thresholding"]
    end

    subgraph RoomNode2["ROOM_NODE_2: Lecture Room 2 (ESP32 #3)"]
        PIR_2["PIR Motion Sensor\n(GPIO 14 - Interrupt)"]
        LDR_2["Ambient Light LDR\n(GPIO 34 - ADC1)"]
        RELAY_2["Lighting Relay / LED\n(GPIO 12)"]
        HB_LED_2["Heartbeat LED\n(GPIO 2)"]
        CTRL_2["Energy Logic: 30s Countdown\n+ Daylight Thresholding"]
```

```
    end

    %% Physical to Tasks
    RFID_IN -->|Swipe Event| Q_ACCESS
    RFID_OUT -->|Swipe Event| Q_ACCESS
    Q_ACCESS --> T_ACCESS
    T_ACCESS -->|Grant/Deny| SERVO_IN
    T_ACCESS -->|Grant/Deny| SERVO_OUT
    T_ACCESS -->|Feedback| LED_GREEN
    T_ACCESS -->|Feedback| LED_RED
    T_ACCESS -->|Lockout/Curfew| BUZZER

    %% Concurrency Interactions
    T_ACCESS <-->|Lock/Verify| M_OCC
    T_ACCESS <-->|Lock/Verify| M_SCHED
    T_ACCESS -->|Push Event| Q_LOG
    Q_LOG --> T_LOG
    T_LOG <-->|Lock/Write| M_SD
    M_SD --> StorageSubsystem

    %% CLI Interactions
    CLI --> AUTH
    AUTH --> T_CLI
    T_CLI <-->|Read Occupants| M_OCC
    T_CLI <-->|Update Window| M_SCHED
    T_CLI <-->|Query Rooms| M_ROOM

    %% Wireless Links
    T_WIFI <--> WIFI_TRANS
    T_HB -->|Ping/Check| WIFI_TRANS
    WIFI_TRANS <--> RoomNode1
    WIFI_TRANS <--> RoomNode2
    RoomNode1 -->|2s Telemetry| WIFI_TRANS
    RoomNode2 -->|2s Telemetry| WIFI_TRANS
    WIFI_TRANS -->|Evacuate Broadcast| RoomNode1
    WIFI_TRANS -->|Evacuate Broadcast| RoomNode2
```

3. Node-by-Node Technical Architecture

3.1 ‘ENTERANCE_MASTER‘ - Entrance Terminal & Gateway

- Physical Ingress/Egress Isolation: The entrance is split into dedicated Entry and Exit corridors. Each corridor has its own RFID reader (RC522 over SPI with dedicated Chip Select pins: GPIO 21 for Entry, GPIO 5 for Exit) and independent servo barrier gates.
- Strict Schedule Curfew (Coursework Section 2.6): When current time is outside authorized hours (e.g., 20:01 to 06:59), the state machine rejects both entry and exit attempts unless the tag possesses admin or security role credentials. Unauthorized exit attempts trigger the buzzer (GPIO 12) and write an AFTER_HOURS_VIOLATION record to access_log.csv.
- Anti-Passback Enforcement: Prevents card sharing. A tag recorded as "Inside" cannot generate an Entry event; a tag recorded as "Outside" cannot generate an Exit event.
- Intrusion Lockout: Three consecutive unregistered or rejected card swipes trigger a 60-second hardware lockout, pulsing the alarm buzzer and preventing brute-force access.
- Single-Writer SD Card Subsystem: Only ENTERANCE_MASTER writes to the MicroSD card. This completely eliminates multi-board race conditions and file lock contentions.

3.2 FreeRTOS Concurrency & Multithreading Layer

To fulfill the requirements of Section 2.7 (Resource Synchronization and System Reliability), ENTERANCE_MASTER employs a multi-threaded FreeRTOS architecture:

```
[Hardware ISRs / Inbound Events]
       RFID Swipe ISR --------> xAccessEventQueue (Depth 10) ------> AccessControlTask (Core 1)
       ESP-NOW Wireless ISR --> xWirelessPacketQueue (Depth 15) ---> WirelessRxTask (Core 1)
       Serial Rx ISR ---------> CLI Parser Buffer -----------------> SerialCLITask (Core 0)
                                                                            |
[Shared State Protection]                                                   |
       xOccupancyMutex -------- Guards in-memory occupancy array <----------+
       xScheduleMutex --------- Guards 07:00-20:00 operating window <-------+
       xRoomStateMutex -------- Guards Room 1 & 2 cached metrics <----------+
       xSDCardMutex ----------- Guards FATFS physical SPI bus <-------------+
                                                                            |
[Non-Blocking File Persistence]                                             |
       AccessControlTask -----> xSDLogQueue (Depth 20) ------------> SDLoggerTask (Core 0)
                                                                            |
                                                                  MicroSD Card (SPI)
```

3.3 ‘ROOM_NODE_1‘ & ‘ROOM_NODE_2‘ - Distributed Lecture Room Controllers

- PIR Motion Sensing & Debounce: Digital input on GPIO 14 triggers an interrupt on motion. A software timer resets a 30-second no-motion countdown. Only when countdown reaches zero does the light relay open.
- Daylight-Aware Illumination (Innovation): An analog LDR photocell on ADC1_CH6 (GPIO 34) measures ambient light. Even if the room is occupied, if ambient light exceeds the threshold (e.g. >250 lux from sunlight), lights remain OFF, saving substantial electricity.
- Autonomous Local Control: Room nodes operate independently. If ENTERANCE_MASTER goes offline, lecture rooms continue sensing motion and switching lights without interruption.
- Wireless Telemetry & Keep-Alive: Every 2 seconds, each room node bundles its status into an ESP-NOW packet:

```
typedef struct __attribute__((packed)) {
    uint8_t  occupied;        // 1 = Occupied, 0 = Vacant
    uint8_t  lights_on;       // 1 = Relay Closed, 0 = Relay Open
    uint16_t light_level_lux; // ADC converted ambient illumination
    uint32_t seconds_active;  // Energy saving counter
} room_status_payload_t;
```

4. Wireless Communication Protocol (ESP-NOW)

Parameter	Specification	Technical Justification
RF Frequency	2.4 GHz ISM Band	Native ESP32 integrated radio; zero external hardware
Protocol Type	ESP-NOW (802.11 Vendor Action Frames)	Connectionless, peer-to-peer; no router, DHCP, or SSID required
Latency	< 4 milliseconds	Immediate response for telemetry streaming and emergency overrides
Addressing	6-Byte Hardware MAC Addressing	Unicast peer registration (ENTERANCE_MASTER, ROOM_NODE_1, ROOM_NODE_2) + Broadcast
Delivery Guarantee	Hardware MAC ACKs + 3x Software Retry	Automatic status reported via esp_now_register_send_cb
Broadcast Channel	FF:FF:FF:FF:FF:FF	Used for zero-latency campus emergency evacuation commands

5. End-to-End Operational Pipelines

Pipeline 1: Student / Staff RFID Ingress

- 1. Tag tapped at Entry Reader $\rightarrow$ SPI interrupt reads UID.
- 2. xAccessEventQueue delivers UID to AccessControlTask.
- 3. Task acquires xScheduleMutex: Verifies current time $\in [07:00, 20:00]$.
- 4. Task verifies users.csv: Tag exists and is authorized.
- 5. Task acquires xOccupancyMutex: Anti-passback check confirms user is currently "Outside".
- 6. User marked "Inside" in memory; xSDLogQueue receives log record (ENTRY, GRANTED).
- 7. GPIO 13 pulses PWM signal $\rightarrow$ Entry servo opens 90 for 5 seconds $\rightarrow$ Green LED blinks.
- 8. SDLoggerTask wakes up, acquires xSDCardMutex, and appends event to access_log.csv.

Pipeline 2: Strict Curfew Lockdown (After-Hours Exit Attempt)

- 1. Tag tapped at Exit Reader at 21:15 $\rightarrow$ UID passed to AccessControlTask.
- 2. Task acquires xScheduleMutex: Current time (21:15) exceeds schedule_end (20:00).
- 3. Task checks user role in users.csv: Role is student (Not Admin).
- 4. Access Denied: Exit servo remains rigidly locked.
- 5. Red LED illuminates; GPIO 12 activates buzzer warning.
- 6. Audit log written to access_log.csv: CURFEW_LOCKDOWN_VIOLATION.

Pipeline 3: Wireless Emergency Evacuation

- 1. Physical button pressed (GPIO 32) OR admin enters emergency on in Serial CLI.
- 2. ENTERANCE_MASTER immediately forces Entry and Exit servos to 90 (unlocked).
- 3. ENTERANCE_MASTER broadcasts MSG_TYPE_EMERGENCY_ALERT over ESP-NOW to FF:FF:FF:FF:FF:FF.
- 4. ROOM_NODE_1 & ROOM_NODE_2 receive broadcast: Override light relays to FORCED ON (illuminating corridors and lecture halls for safe evacuation).
- 5. ENTERANCE_MASTER flushes occupancy.csv and logs CAMPUS_EMERGENCY_EVACUATION.